Simple Linear Regression: Philippines Population

(2010 - 2025, Least Squares Fit)

Year	Population (persons)
2010	94,636,700
2011	96,337,913
2012	98,032,317
2013	99,700,107
2014	101,325,201
2015	103,031,365
2016	104,875,266
2017	106,738,501
2018	108,568,836
2019	110,380,804
2020	112,190,977
2021	113,880,328
2022	115,559,009
2023	117,337,368
2024	119,106,224
2025	120,864,358

Regression Equation: $y = -3{,}442{,}307{,}421.96 + 1{,}759{,}587.49 \cdot x$

Slope a_1 = 1,759,587.49 persons/year

Intercept a_0 = -3,442,307,421.96

S_r (SSE) = 183,554,564,467.77

r^2 = 0.999826

Standard error $s_{y/x}$ = 114,503.45 persons

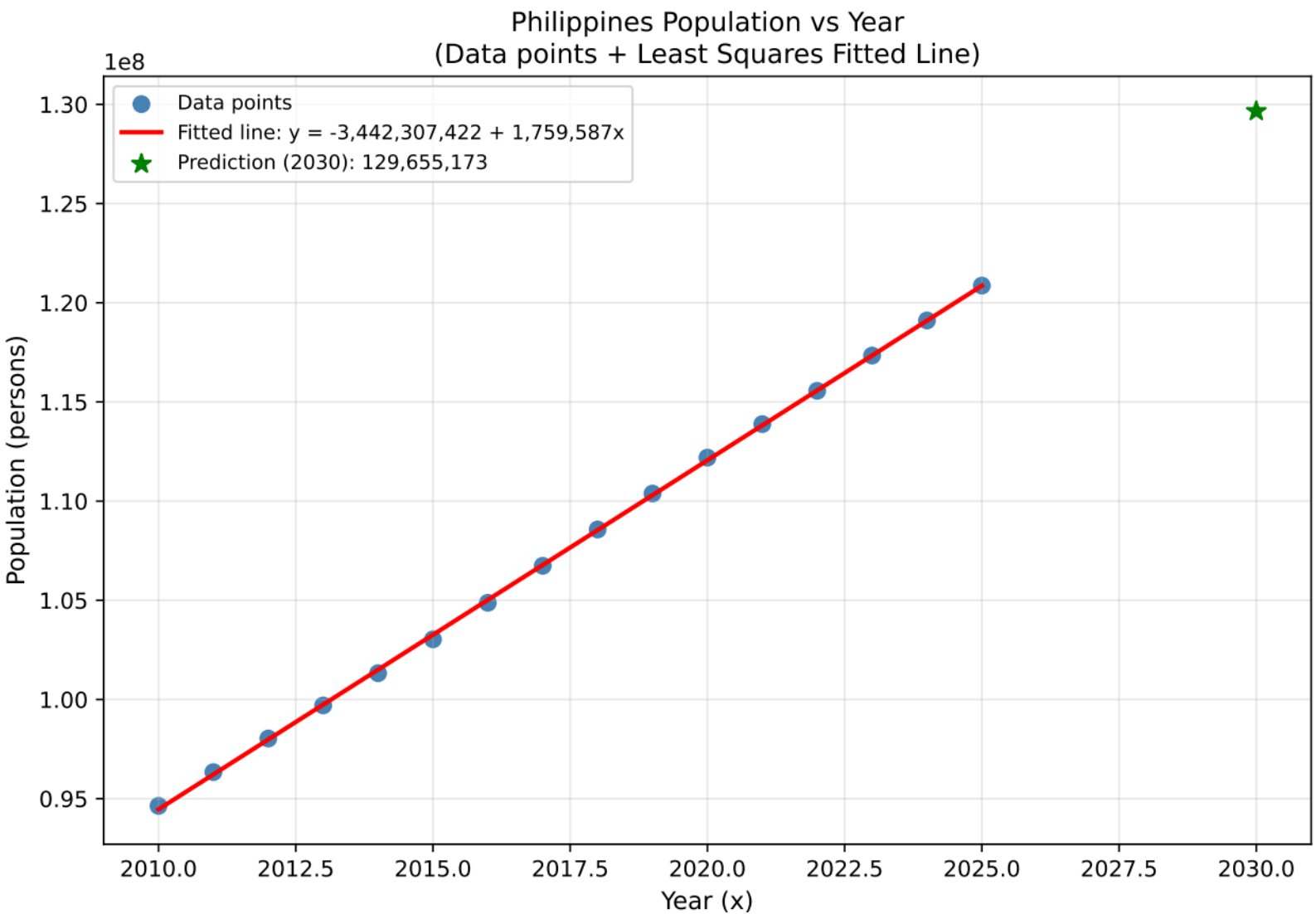
Prediction at x = 2030: 129,655,173 persons

Source: Macrotrends Philippines Population Tracker referenced from Philippine Statistics Authority

URL: https://www.macrotrends.net/global-metrics/countries/phl/philippines/population

The data represent the total population of the Philippines from 2010 to 2025, measured in persons. The dataset contains 16 paired observations, with each observation consisting of a year and its corresponding population.

The purpose of the analysis is to determine the relationship between time and population using simple linear regression. The regression model estimates how much the population changes per year and can be used to predict the population in a future year


Philippines Population vs Year
(Data points + Least Squares Fitted Line)
1e8
Data points
Fitted line: y = -3,442,307,422 + 1,759,587x
Prediction (2030): 129,655,173
Population (persons)
Year (x)
1.30
1.25
1.20
1.15
1.10
1.05
1.00
0.95
2010.0
2012.5
2015.0
2017.5
2020.0
2022.5
2025.0
2027.5
2030.0

Dependent variable (x): Year, measured in calendar years (2010–2025).

Dependent variable (y): Total population of the Philippines, measured in persons.

Interpretation:

For every one-year increase in time, the regression model estimates that the population of the Philippines increases by approximately 1,759,587 persons per year, on average.

For example, the model estimates that the population in 2021 would be approximately 1.76 million higher than the predicted population in 2020.

This is an average trend estimated from the data, not an assertion that the population increases by exactly this number every year

Mathematically, the intercept represents the population predicted by the regression equation when $x = 0$, or calendar year 0.

However, this value is negative and does not have a meaningful real-world interpretation for the population of the Philippines. Year 0 is far outside the dataset's range of 2010–2025.

Therefore, the intercept is useful for defining the regression line mathematically, but it should not be interpreted as an actual historical population.

Interpretation of R^2

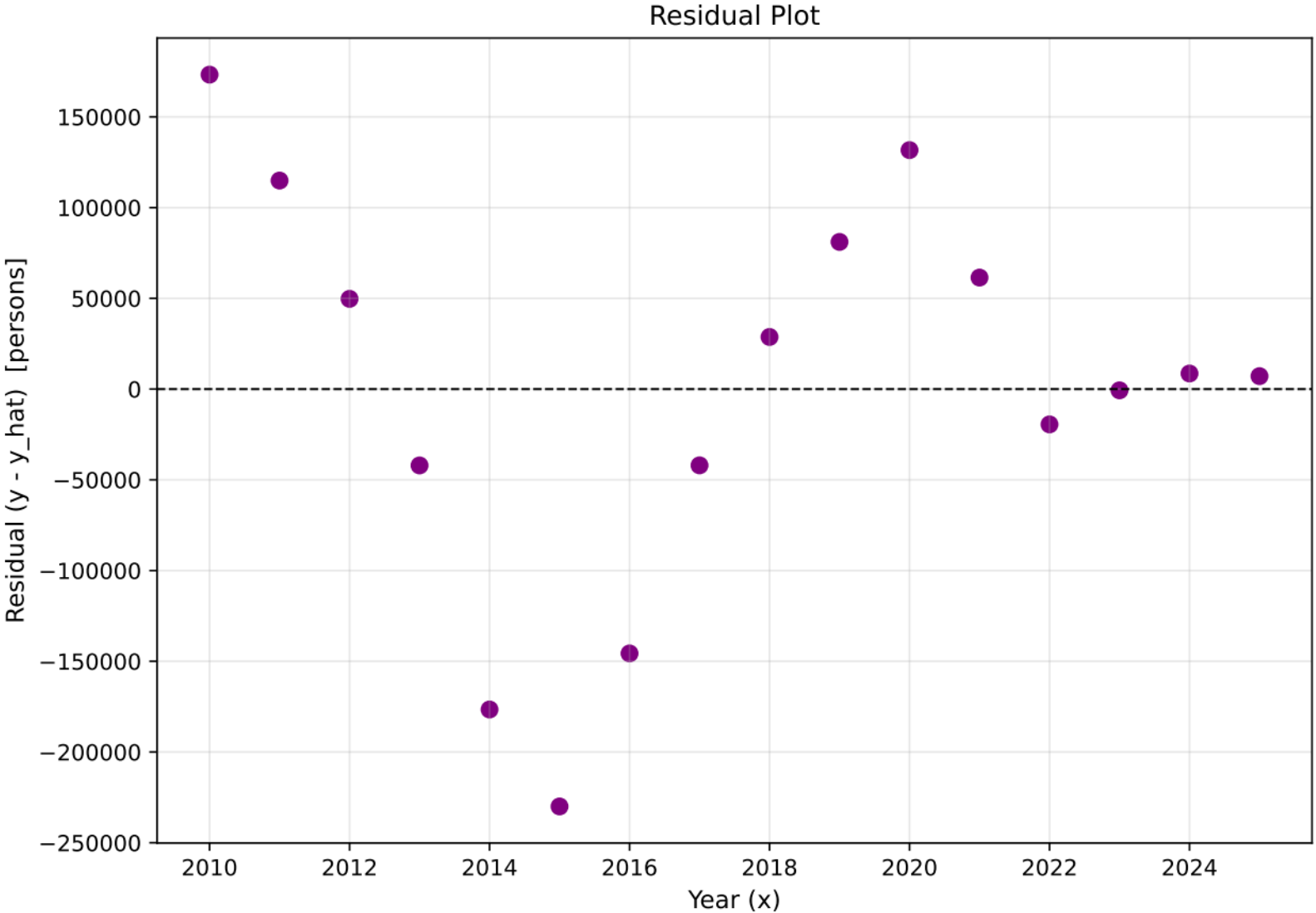
The R^2 The value of 0.999826 means that approximately 99.9826% of the variation in population within this dataset is explained by its linear relationship with year.

This indicates an extremely strong linear fit over the observed period.

Interpretation of the standard error

The standard error of 114,503.45 persons measures the typical size of the regression's residuals, expressed in population units. The actual population values typically deviate from the fitted values by around 114,503 persons.

Relative to a population exceeding 94 million throughout the dataset, this is a small error, supporting the conclusion that the fitted line closely follows the observations.


Residual Plot
150000
100000
50000
0
−50000
−100000
−150000
−200000
−250000
Residual (y - y_hat) [persons]
2010
2012
2014
2016
2018
2020
2022
2024
Year (x)

The residual plot shows that the residuals are relatively close to zero compared with the population values. However, they are not randomly scattered perfectly: they start positive, become negative during the middle years, and return to positive values toward the end.

The prediction provides an estimate of the Philippine population in 2030 based on the observed upward trend from 2010 to 2025. Since the prediction is only five years beyond the available data, it can be useful as a simple estimate for preliminary planning and statistical analysis.

However, it assumes that the historical linear trend continues. Changes in birth rates, death rates, migration, and other demographic factors may cause the actual population to differ from the prediction. It should therefore be treated as a model-based estimate, not an official population forecast.